

Multigenerational Mobility and Long-Run Inequality

Taylan Alpkaya¹ **Yonatan Berman**² Ravi Kanbur³

¹University of Mannheim

²King's College London, LSE & IZA

³Cornell University

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A problem with intergenerational mobility measurement

- ▶ Since the 1970s, mobility is often summarized by one slope:

$$y_{i,t} = \alpha + \beta y_{i,t-1} + \varepsilon_{i,t}$$

- ▶ Simple, useful, and widely used

A problem with intergenerational mobility measurement

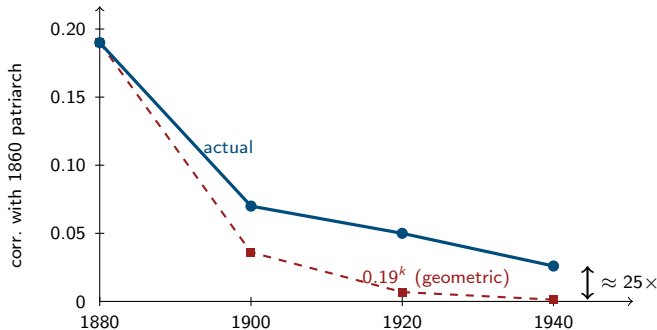
- ▶ Since the 1970s, mobility is often summarized by one slope:

$$y_{i,t} = \alpha + \beta y_{i,t-1} + \varepsilon_{i,t}$$

- ▶ Simple, useful, and widely used
- ▶ However, $y_{i,t} = \alpha + \beta y_{i,t-1} + \varepsilon_{i,t}$ also implicitly predicts:
 1. Fast decay of multigenerational correlations (no long-run persistence)
 2. Great Gatsby curve not well replicated
- ▶ Can we improve?
- ▶ Why should we care?

A problem with intergenerational mobility measurement

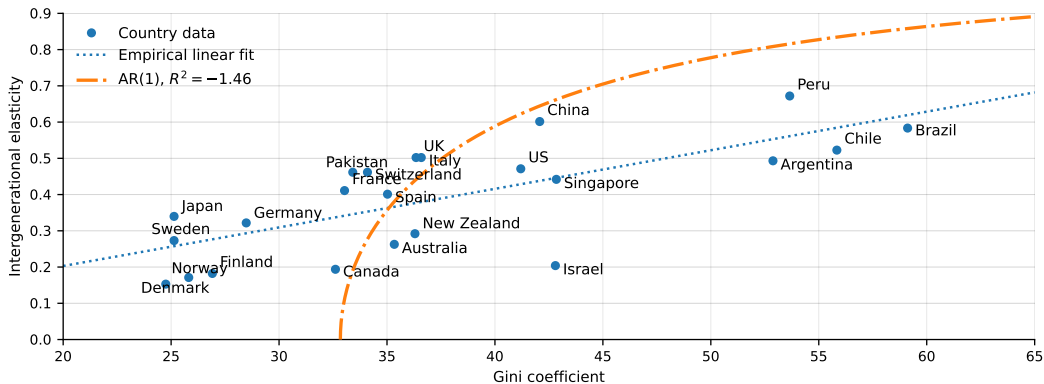
- Under geometric decay, correlations vanish within two–three generations (for historical US Census data)



1860 correlation = 1 by construction; in 1940: actual 0.026 vs. 0.001 implied. Same pattern across countries and proxies (Clark, 2014; Lindahl *et al.*, 2015; Braun & Stuhler, 2018; Nybom & Stuhler, 2024)

A problem with intergenerational mobility measurement

► Also, bad fit to cross-country Great Gatsby curve



Data from Corak (2013)

A problem with intergenerational mobility measurement

- ▶ Luckily – multigenerational data are increasingly available (although still rare):
 - ▶ US Census Linking Project
 - ▶ Digitized historical archives (Florence; England)
 - ▶ Swedish Multi-generation Register
 - ▶ Time...
- ▶ We should be able to improve our model

This paper

- ▶ A minimal fix to AR(1): a permanent lineage component θ_g – AR(1)+FE, identified with one additional generation
- ▶ Tested against AR(1), AR(2), $\dots$, AR(p) benchmarks in linked US censuses, 1850–1940 – 1.6M observations, 560K dynasties:
 - ▶ in-sample goodness of fit
 - ▶ out-of-sample prediction of inequality and persistence
- ▶ Establishing it as the better model – closed-form long-run inequality and persistence

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 - ▶ in-sample goodness of fit
 - ▶ out-of-sample prediction of inequality and persistence
- ▶ Establishing it as the better model – closed-form long-run inequality and persistence
- ▶ Key insight: the dispersion of θ_g , not the elasticity β , is the main driver of both inequality and persistence

Related literature

- ▶ Classic intergenerational mobility: Becker–Tomes (1979); Solon (1992); Chetty *et al.* (2014); the Great Gatsby curve (Krueger, 2012; Corak, 2013)
- ▶ Multigenerational persistence: Stuhler (2012); Clark (2014); Lindahl *et al.* (2015); Braun & Stuhler (2018); Solon (2018); Belloc *et al.* (2024)
- ▶ Identifying persistent components: latent factors and kin (Vosters & Nybom, 2017; Adermon *et al.*, 2021; Collado *et al.*, 2023; Nybom & Stuhler, 2024); surnames (Güell *et al.*, 2015; Santavirta & Stuhler, 2024)
- ▶ Long-run implications of mobility processes: Solon (2004); Becker *et al.* (2018); Nybom & Stuhler (2024)
- ▶ US mobility in linked historical censuses: Abramitzky *et al.* (2021); Ward (2023); Jácome *et al.* (2025)

Contribution

1. Mobility models evaluated as forecasts, not only in-sample fit. If a model is good – it would ideally predict future inequality and persistence
2. Mapping model parameters to long-run inequality and persistence
3. Structural Great Gatsby curve

The model: AR(1) + dynasty fixed effects

- ▶ A time-invariant lineage component θ_g in the law of motion:

$$y_{i,g,t} = \alpha + \beta y_{i,g,t-1} + \theta_g + \varepsilon_{i,g,t}$$

- ▶ $y_{i,g,t}$: outcome (individual i ; dynasty g ; generation t)
- ▶ β : within-lineage reversion
- ▶ σ_θ : dispersion of permanent lineage means
- ▶ σ_ε : transitory shocks

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- ▶ β : within-lineage reversion
- ▶ σ_θ : dispersion of permanent lineage means
- ▶ σ_ε : transitory shocks
- ▶ Nests AR(1) at $\sigma_\theta = 0$
- ▶ Pooled OLS slope $\neq \beta$ (conflates β with σ_θ^2 (Stuhler, 2012))
- ▶ No geometric decay

Data: linked full-count US censuses

- ▶ IPUMS full-count censuses (Ruggles *et al.*, 2024)
 - ▶ > 800M person records, 1850–1940
- ▶ **Census Tree** links (Price *et al.*, 2021; Buckles *et al.*, 2024)
 - ▶ FamilySearch genealogies + machine learning + existing linking projects
- ▶ Nine waves: 1850/1860/1870/1880/1900/1910/1920/1930/1940 – up to five generations

Sample selection

- ▶ Start with ancestor aged 30–40 in the initial wave
- ▶ Link descendants if observed aged 30–40 in *any* future census
- ▶ A **dynasty**: descendants of a common patriarch
- ▶ We keep only dynasties of 3 generations and above

Sample	Size
Father–son pairs (≥ 3 gen)	10.6M
Dynasties (≥ 3 gen)	2.8M
5-generation subsample dynasties	27,442 18,850
Key sample dynasties	1.6M 560k

Outcome: the occupational income score

- ▶ Each occupation = its workers' median income in the 1950 census
 - ▶ IPUMS OCCSCORE on OCC1950 codes; hundreds of 1950 dollars
- ▶ Earnings unobserved before 1940 – one proxy, consistent across waves
- ▶ Measures between-occupation differences, not individual earnings
- ▶ Mean normalized across waves $\Rightarrow$ relative standing
- ▶ Workhorse of US historical mobility
 - ▶ Olivetti, Paserman & Salisbury (2018); Abramitzky *et al.* (2021); Ward (2023)

Testing the model

- ▶ Comparing $AR(1)+FE$ to $AR(1), \dots, AR(p)$ and to placebo FE model
- ▶ **In sample:**
 - ▶ Does adding dynasty fixed effects better explain data?
 - ▶ Placebo FE – is the FE gain mechanical?
- ▶ **Out of sample:**
 - ▶ Estimate models using generations 1–3; then predict generations 4 and 5 (and combinations)
 - ▶ Which model best predicts persistence?
 - ▶ Which model best predicts inequality?

In sample: Dynasties matter, not only number of degrees of freedom

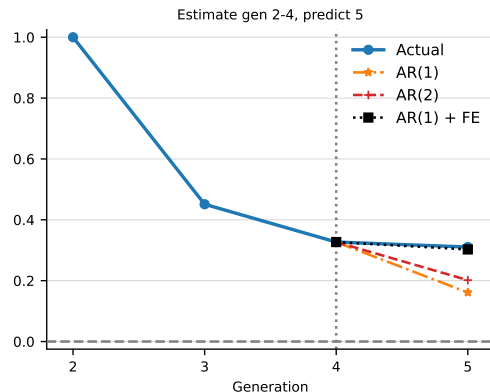
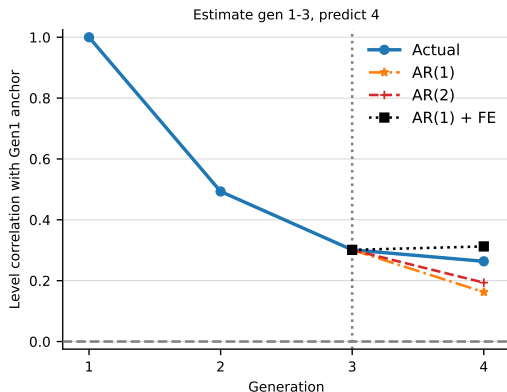
	AR(1)	AR(2)	Placebo FE	FE
Father	0.4914*** (0.0010)	0.4048*** (0.0011)	0.4916*** (0.0010)	0.2377*** (0.0023)
Grandfather		0.1919*** (0.0012)		
R^2	0.2037	0.2239	0.4702	0.5868
Adj. R^2	0.2037	0.2239	0.2036	0.3789
ΔR^2		0.0202	—	0.1166
Observations (N)	1,550,634	1,550,634	1,550,634	1,550,634
Dynasties	519,102	519,102	519,102	519,102

Out of sample: predicting long-run persistence and inequality

- ▶ Estimate models on generations 1–3; forecast generations 4 and 5 (and combinations) for same dynasties
- ▶ Forecasting uses Monte-Carlo: simulate noise and estimate outcome
- ▶ Two targets: the correlation with the dynasty patriarch; the cross-sectional Gini
- ▶ Standard in macro/finance (Stock & Watson, 2003; Welch & Goyal, 2008); new to mobility

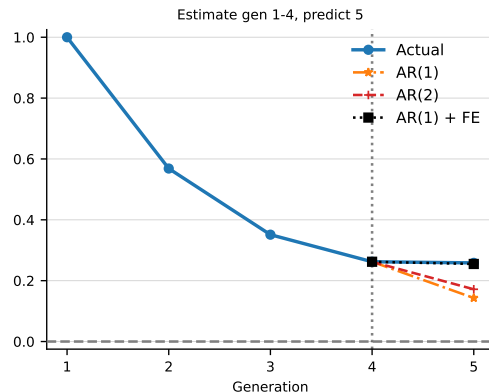
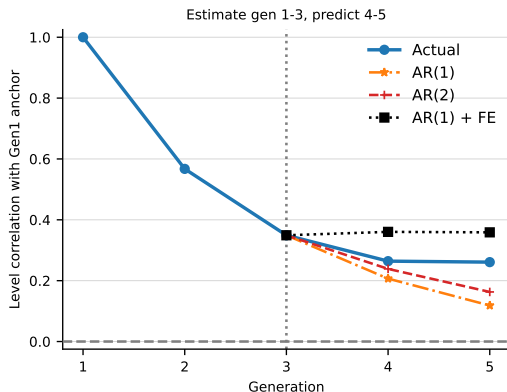
Out of sample I: persistence

- AR(1)+FE prediction of the actual correlation is better than AR(1) and AR(2), which quickly decay to zero



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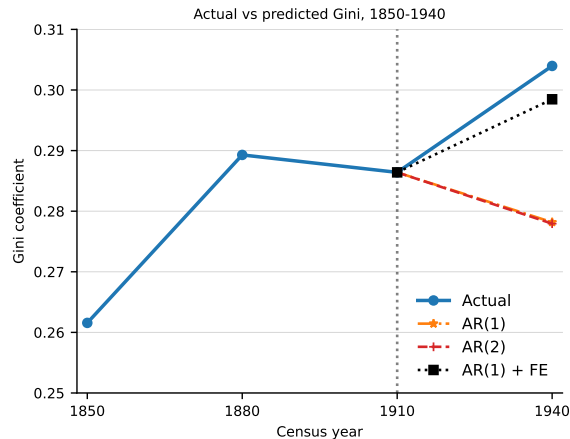
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Out of sample II: inequality

- ▶ Actual Gini: 0.286 (1910) $\rightarrow$ 0.304 (1940)
- ▶ AR(1) and AR(2) compress the distribution – mispredicts the rise
- ▶ AR(1)+FE: fixed effects counteract the compression

Gini	1940
Actual	0.304
AR(1)	0.278
AR(2)	0.278
AR(1)+FE	0.298



Test summary

- ▶ AR(1)+FE is a better model than AR(p)
- ▶ Key idea: β **reverts**, θ_g **persists**
- ▶ Adding more lags does not solve the problem – a permanent component is needed

	AR(1)	AR(2)	AR(1)+FE
In-sample goodness-of-fit	low	low	higher
Long-run persistence	vanishes	vanishes	matches
Inequality prediction quality	low	low	high

Implications I: why should we care?

- ▶ So far, AR(1)+FE wins on statistical grounds (if we can estimate it)
- ▶ It better explains the data, and identifies β more cleanly

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Implications I: why should we care?

- ▶ So far, AR(1)+FE wins on statistical grounds (if we can estimate it)
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- ▶ Good per se – but not the main point
- ▶ What we really want: its implications for long-run persistence and long-run inequality
- ▶ To get there, we study the model in **steady state**

Implications II: the model in steady state

- ▶ The model is a *law of motion* for the outcome:

$$y_{i,g,t} = \alpha + \underbrace{\beta}_{\text{reversion}} y_{i,g,t-1} + \underbrace{\theta_g}_{\substack{\text{lineage,} \\ \text{var } \sigma_\theta^2}} + \underbrace{\varepsilon_{i,g,t}}_{\substack{\text{shock,} \\ \text{var } \sigma_\varepsilon^2}}$$

- ▶ Three primitives: β (within-lineage reversion), σ_θ (lineage dispersion), σ_ε (transitory shocks)
- ▶ Iterate forward $\Rightarrow$ converges to a steady state; with Gaussian shocks, in closed form
- ▶ Three questions it lets us answer:
 1. How fast do we get there? (is the steady state relevant?)
 2. How much **inequality** does it imply?
 3. How much **long-run persistence**?

Implications III: in the long run, income is log-normal

- ▶ Treat $y_{i,g,t}$ as log-income (as usual). With Gaussian shocks, $\log y$ is a sum of normals
 - ▶ normal at every generation; in the limit, **log-normal**
- ▶ A log-normal is determined by its log-variance σ_∞^2
- ▶ So steady-state inequality is **exact**: the Gini is a closed form in σ_∞^2

$$\text{Gini}_\infty = \text{erf}\left(\frac{\sigma_\infty}{2}\right)$$

Implications IV: two sources of long-run inequality

- ▶ That log-variance splits into a **permanent** and a **transitory** part:

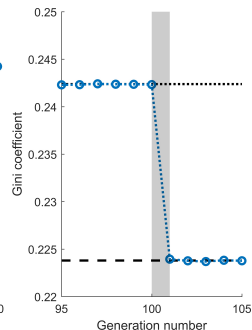
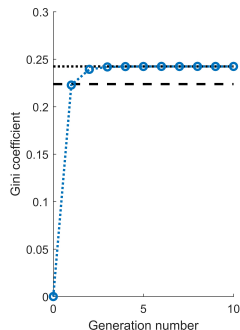
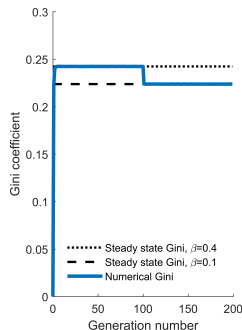
$$\sigma_{\infty}^2 = \underbrace{\frac{\sigma_{\theta}^2}{(1-\beta)^2}}_{A \text{ (lineage)}} + \underbrace{\frac{\sigma_{\varepsilon}^2}{1-\beta^2}}_{B \text{ (shocks)}}$$

- ▶ Long-run persistence (corr. of generation 1 with generation t) converges to the permanent share: $\rho_{S,\infty} \propto A/(A+B)$
- ▶ The closed forms let us ask: which primitive – β , σ_{θ} , or σ_{ε} – matters most for inequality and persistence?
- ▶ We answer with **elasticities** – % change in inequality per % change in a primitive:

$$\eta_x = \frac{\partial \log \text{Gini}_{\infty}}{\partial \log x}, \quad x \in \{\beta, \sigma_{\theta}, \sigma_{\varepsilon}\}$$

Result 1: the steady state arrives fast

- First, need to establish that the steady state is justified
- Simulate the model forward from equality
- All specifications settle within **2–3 generations**
- The steady state is not a distant limit – a distribution reached quickly

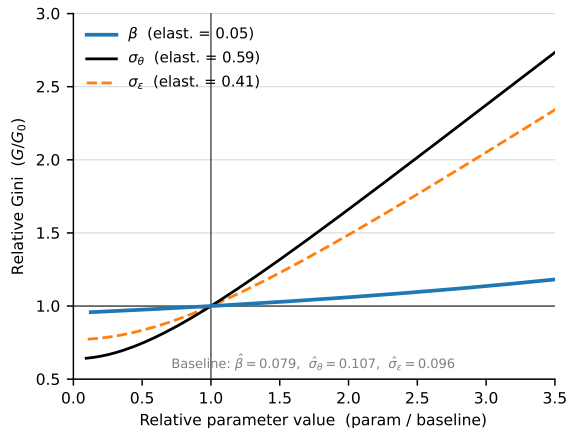


Result 2: what drives steady-state inequality?

- ▶ Steady-state inequality rises with all three primitives – but not equally
- ▶ Elasticities at the 1940 estimates:

$$\underbrace{\eta_{\sigma_{\theta}}}_{0.59} > \underbrace{\eta_{\sigma_{\varepsilon}}}_{0.41} > \underbrace{\eta_{\beta}}_{0.05}$$

- ▶ **Lineage dispersion dominates**; the parent–child slope is weakest
- ▶ Halving β barely moves inequality; halving σ_{θ} does

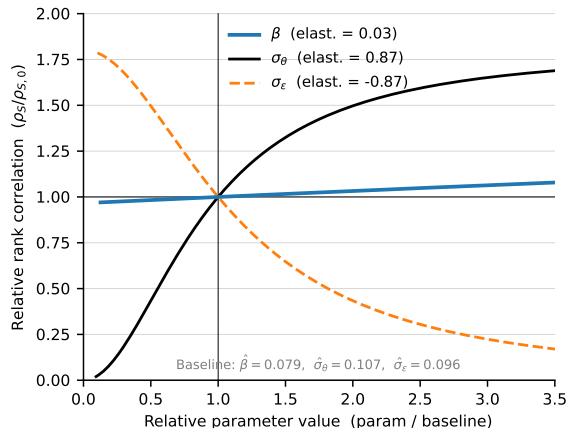


Result 3: long-run persistence

- Same ordering by magnitude – σ_θ dominates:

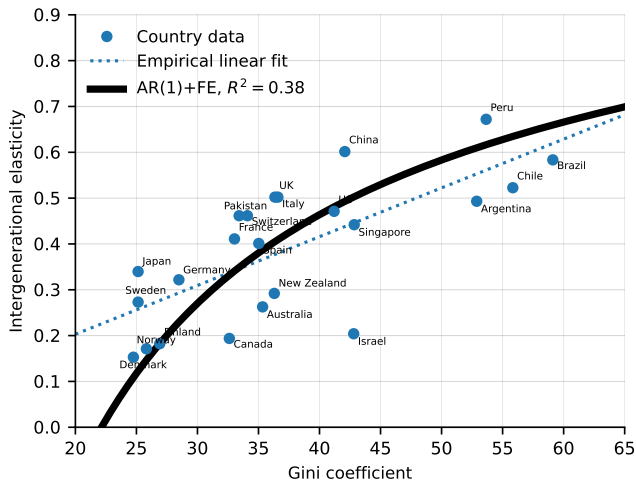
$$|\eta_{\sigma_\theta}^\rho| > |\eta_{\sigma_\varepsilon}^\rho| > |\eta_\beta^\rho|$$

- But the **sign** differs: $\sigma_\theta \uparrow$ raises persistence, $\sigma_\varepsilon \uparrow$ **lowers** it
- Both raise inequality – so only σ_ε **trades** inequality against persistence



Result 4: a structural Great Gatsby curve

- ▶ If σ_θ varies alone (hold $\beta, \sigma_\varepsilon$ fixed): inequality and immobility rise together
- ▶ This creates a **structural Great Gatsby curve** – increasing and convex
- ▶ It fits the data better than a linear curve, and better than the AR(1) and AR(2) fit



Conclusion

1. The lineage fixed effects dispersion σ_θ dominates long-run inequality and long-run persistence; β is a weaker primitive
2. σ_θ and σ_ε both raise inequality but move persistence in opposite directions – only σ_ε trades the two off
3. We reproduce the **Great Gatsby curve** as a structural, testable object

► Next steps (not this paper):

- β and θ_g identified separately – so: which policies move which?
- β ? universal public education, school quality, credit for the young
- σ_θ ? wealth and inheritance taxes, place-based policy, desegregation, reparations
- Not mutually exclusive – but potentially bigger bang-for-buck from σ_θ -policies in the long run

Conclusion

- ▶ Our statistics are still mostly two-generation autoregressions – we can do better
- ▶ AR(1)+FE – one permanent lineage component – **fits and forecasts better** than AR(p)
- ▶ At the estimates:
 - ▶ lineage dispersion σ_θ drives long-run inequality and persistence
 - ▶ the parent–child slope β is a weaker primitive
- ▶ **Reframing – two mechanisms, not one:**
 - ▶ intergenerational reversion (β) and inherited advantage (σ_θ) – usually conflated
 - ▶ they affect long-run inequality and persistence differently
 - ▶ and different policies may move each

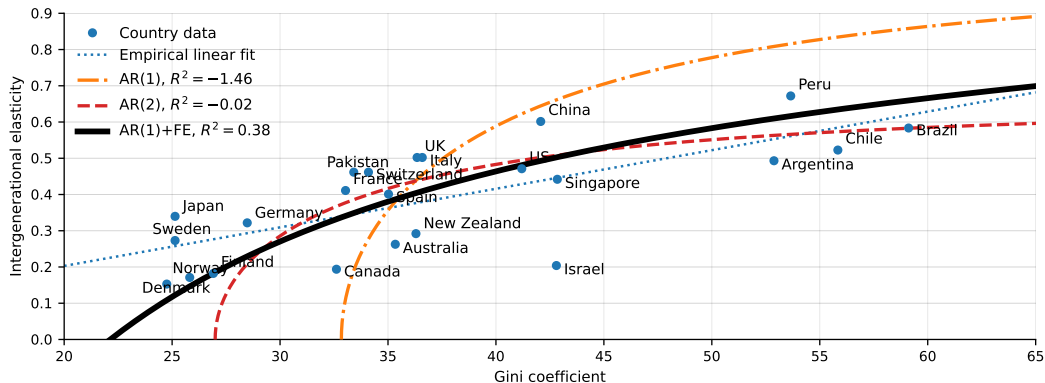
Backup – relation to latent-factor models

- ▶ Latent-factor (Vosters & Nybom, 2017; Collado, Ortuño-Ortín & Stuhler, 2023; Nybom & Stuhler, 2024):

$$a_{i,g,t} = \lambda a_{i,g,t-1} + u_{i,g,t}, \quad y_{i,g,t} = \rho a_{i,g,t} + e_{i,g,t}$$

- ▶ identify λ through the loading ρ : a measurement model on proxies, or kin of varying genetic distance
- ▶ AR(1)+FE identifies θ_g from **depth** (≥ 3 generations) + **breadth** (brothers, cousins), with no measurement model
- ▶ Complementary, not rival: latent-factor suits *limited depth, rich kin*; ours suits *many generations, many dynasties*, and yields a falsifiable closed-form steady state

A structural Great Gatsby curve comparison



Data from Corak (2013)

In sample: OCCSCORE ranks

	AR(1)	AR(2)	AR(3)	AR(4)	Dynasty FE	Placebo FE
Father	0.240*** (0.000)	0.233*** (0.001)	0.199*** (0.002)	0.193*** (0.007)	0.237*** (0.001)	0.241*** (0.000)
Grandfather		0.079*** (0.0005)	0.089*** (0.0019)	0.040*** (0.0075)		
Great-grandfather			0.031*** (0.0018)	0.034*** (0.0084)		
Great ² -grandfather				0.011 (0.0072)		
Observations	10.6M	5.2M	468k	27k	10.6M	10.6M
Adj. R^2	0.052	0.061	0.054	0.044	0.235	0.054
Within R^2					0.027	0.053

Percentile ranks (0–100); dynasties with ≥ 3 generations. SE clustered at the dynasty level. * $p < .1$, ** $p < .05$, *** $p < .01$.

In sample: Dynasties matter, not only number of degrees of freedom

	AR(1)	AR(2)	Placebo FE	FE	+Sib	+Unc	+Cous	+GF
Father	0.4914*** (0.0010)	0.4048*** (0.0011)	0.4916*** (0.0010)	0.2377*** (0.0023)	0.2987*** (0.0009)	0.2362*** (0.0010)	0.2258*** (0.0010)	0.2114*** (0.0010)
Grandfather		0.1919*** (0.0012)					0.0705*** (0.0009)	
Sibling mean					0.3903*** (0.0012)	0.3595*** (0.0012)	0.3464*** (0.0013)	0.3408*** (0.0013)
Uncle mean						0.1612*** (0.0010)	0.1159*** (0.0011)	0.0915*** (0.0011)
Cousin mean							0.1018*** (0.0010)	0.0944*** (0.0010)
R^2	0.2037	0.2239	0.4702	0.5868	0.2958	0.3072	0.3122	0.3144
Adj. R^2	0.2037	0.2239	0.2036	0.3789	0.2958	0.3072	0.3122	0.3144
ΔR^2		0.0202	—	0.1166	0.0921	0.1035	0.1084	0.1107
Observations (N)	1,550,634	1,550,634	1,550,634	1,550,634	1,550,634	1,550,634	1,550,634	1,550,634
Dynasties	519,102	519,102	519,102	519,102	519,102	519,102	519,102	519,102

Backup – correlates of dynasty fixed effects

Covariate	Pooled
Share urban	13.54*** (0.48)
Share farmer	-1.10** (0.47)
Share Black	-22.61*** (0.84)
Share Northeast	-14.49*** (0.24)
Share South	-20.80*** (0.25)
Share West	0.33 (0.55)
Share Immigrated from Father's State	1.59*** (0.54)
Share Immigrated from Founder's State	0.51* (0.27)
Mean censuses linked	3.84*** (0.22)
Within-gen. SD of <i>songscore</i>	0.27*** (0.01)
Log dynasty size	-4.03*** (0.28)
Mean sons per member	-4.52*** (0.24)
R^2	0.466